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STRUCTURAL ASSEMBLY HAVING POCKETS

Abstract

A structural assembly for a vehicle includes a cast body having a main wall, a non-circular shaped pocket formed in the main wall, and at least one mounting feature. The pocket is defined at least partially by a base wall and an arcuate wall. The base wall has a first end and a second opposing end. The arcuate wall extends in a first direction from the first end of the base wall to the main wall. The arcuate wall also extends from a first side of the pocket to a second opposing side of the pocket. The mounting feature extends in a second direction from the base wall that is opposite the first direction and is configured to be coupled to a vehicle component.

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Background/Summary

FIELD

[0001] The present disclosure relates to a structural assembly having pockets.

BACKGROUND

[0002] The statements in this section merely provide background information related to the present disclosure and does not constitute prior art.

[0003] Vehicle structures may be manufactured by a casting process including filling molten or semisolid material into a mold and then hardening the material into a shape defined by the mold. The vehicle structure manufactured by the casting process may include features that allow the structure to interact (e.g., be secured) to other components or parts of the vehicle.

[0004] The present disclosure addresses the mechanical properties of the structural castings, which may include features that interact to other components or parts of the vehicle.

SUMMARY

[0005] This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

[0006] In one form, the present disclosure provides a structural assembly for a vehicle that includes a cast body. The cast body includes a main wall, a non-circular shaped first pocket formed in the main wall, and at least one mounting feature. The first pocket is defined at least partially by a base wall and an arcuate wall. The base wall has a first end and a second opposing end. The arcuate wall extends in a first direction from the first end of the base wall to the main wall. The arcuate wall also extends from a first side of the first pocket to a second opposing side of the first pocket. The mounting feature extends in a second direction from the base wall that is opposite the first direction. The feature is also configured to be coupled to a vehicle component.

[0007] In variations of the structural assembly of the above paragraph, which can be implemented individually or in any combination: the mounting feature includes a plurality of mounting features extending in the second direction from the base wall; the mounting feature is a boss, the boss comprises an aperture extending at least partially therethrough; the boss comprises a length having a variable thickness; the boss has a proximal end and a distal end, the proximal end has a greater thickness than a thickness of the distal end; the first pocket has a Kammtail virtual foil profile; the base wall is planar; the cast main body further includes a non-circular shaped second pocket, the second pocket has a Kammtail virtual foil profile; the first pocket extends in a third direction and the second pocket extends in a fourth direction, the third direction different from the fourth direction; the first pocket extends perpendicular to the mounting feature; and the cast main body is a front-end structure of the vehicle, the front end structure includes a left upper rail and a right upper rail.

[0008] In another form, the present disclosure provides a structural assembly for a vehicle that includes a cast body. The cast body includes a main wall, a non-circular shaped first pocket formed in the main wall, and a plurality of bosses. The first pocket is defined at least partially by a base wall and an arcuate wall. The base wall has a first end and a second opposing end that is straight. The arcuate wall extends in a first direction from the first end of the base wall to the main wall. The arcuate wall also extends from a first side of the first pocket to a second opposing side of the first pocket. Each boss of the plurality of bosses extend in a second direction from the base wall and comprises an aperture extending at least partially therethrough. The second direction is opposite the first direction. Each boss of the plurality of bosses is configured to be coupled to a vehicle component.

[0009] In variations of the structural assembly of the above paragraph, which can be implemented individually or in any combination: each boss has a proximal end and a distal end, the proximal end has a greater thickness than a thickness of the distal end; the first pocket has a Kammtail virtual foil profile; the cast body further includes a non-circular shaped second pocket, the second pocket has a Kammtail virtual foil profile; the first pocket extends in a third direction and the second pocket

extends in a fourth direction, the third direction different from the fourth direction; the cast body is a front-end structure of the vehicle, the front end structure includes a left upper rail and a right upper rail; and the main wall is inclined.
[0010] In yet another form, the present disclosure provides a structural assembly for a vehicle that includes a cast body. The cast body includes a main wall, a plurality of non-circular shaped pockets formed in the main wall, and a plurality of bosses. Each pocket is defined at least partially by a base wall and an arcuate wall. The base wall has a first end and a second opposing end that is straight. The arcuate wall extends in a first direction from the first end of the base wall to the main wall. The arcuate wall also extends from a first side of the first pocket to a second opposing side of the first pocket. Each boss of the plurality of bosses extend in a second direction from the base wall that is opposite the first direction. Each boss is configured to be coupled to a vehicle component and includes a proximal end and a distal end. The proximal end has a greater thickness than a thickness of the distal end. A first non-circular shaped pocket of the plurality of non-circular shaped pockets extends in a third direction and a second non-circular shaped pocket of the plurality of non-circular shaped pockets extends in a fourth direction. The third direction different from the fourth direction.
[0011] In variations of the structural assembly of the above paragraph, each non-circular shaped pocket of the plurality of non-circular shaped pockets has a Kammtail virtual foil profile.
[0012] Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS
[0013] In order that the disclosure may be well understood, there will now be described various forms thereof, given by way of example, reference being made to the accompanying drawings, in which:
[0014] FIG. **1** is a perspective view of a portion of a vehicle body including a structural assembly having a casting according to the principles of the present disclosure;
[0015] FIG. **2** is a perspective view of the casting of the structural assembly of FIG. **1**;
[0016] FIG. **3** is a top view of the casting of the structural assembly of FIG. **1**;
[0017] FIG. **4**A is a perspective bottom view of the casting of the structural assembly of FIG. **1**;
[0018] FIG. **4**B is a perspective view of a portion of the structural assembly indicated as **4**B in FIG. **4**A;
[0019] FIG. **5** is a perspective view of a portion of the casting of the structural assembly of FIG. **1**;
[0020] FIG. **6** is a cross-sectional view of the casting of the structural assembly of FIG. **5** taken along line **6-6**;
[0021] FIG. **7** is another cross-sectional view of the casting of the structural assembly of FIG. **1** through one of the pockets;
[0022] FIG. **8** is a schematic view illustrating molten flowing through a mold according to the principles of the present disclosure; and
[0023] FIG. **9** is a flow chart illustrating a method of forming a structural assembly according to the principles of the present disclosure.
[0024] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.
DETAILED DESCRIPTION
[0025] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses. It should be understood that throughout the drawings,

corresponding reference numerals indicate like or corresponding parts and features.

[0026] Referring to FIG. **1**, a vehicle **12** including a structural assembly **10** is provided. In the example illustrated, the vehicle **12** is of a uni-body construction. In some forms, the vehicle **12** may be a body-on-frame architecture. In some forms, the vehicle **12** can be an electric vehicle. In other forms, the vehicle **12** can have an internal combustion engine (ICE). Still, in other forms, the vehicle **12** can be a hybrid electric vehicle (HEV). The structural assembly **10** according to the present disclosure includes a casting or cast main body **14** and a pair of strut caps (not shown) secured to the casting **14**.

[0027] A "casting" **14** as used herein is a product that is manufactured by a casting process, i.e., by filling molten or semisolid material into a mold and then hardening the material into a shape defined by the mold. The casting **14** includes a plurality of structural components joined together in the casting process. The plurality of structural components form structural load paths and extend in a plurality of directions in a 3D space. That is, the casting process providing the casting **14** with modes of deformation and energy absorption throughout 3D space that absorbs energy more readily than forming each component separately and joining them later. By forming the casting **14** with the casting process, the casting **14** increases uniformity of energy absorption characteristics throughout the structural components and reducing overall weight of portions of the structural assembly **10**. The casting process reduces machining of the structural components upon forming the casting **14**, reducing manufacturing time.

[0028] The casting **14** is formed of a cast-allowable alloy material, i.e., a material that is able to be used in the casting process. As described above, the material used in the casting process is heated to a liquid or semisolid phase and then introduced to the mold to form the casting **14**. The cast-allowable alloy material can be, e.g., one of an aluminum alloy, a fiber-reinforced aluminum, a fiber-reinforced magnesium, etc. The cast-allowable alloy material may have a lower weight, ductility, and/or toughness than other materials used in vehicle components. That is, the cast-allowable alloy material may be less ductile and may have a lower toughness than other materials that are not able to be used in the casting process. The lower weight of the cast-allowable alloy reduces overall weight of the vehicle **12**.

[0029] Referring to FIGS. **2**, **3**, **4**A, the casting **14** may be of a front end of the vehicle **12** and may be secured to a subframe (not shown) of the vehicle **12** such that the subframe is supported by the casting **14**. In the example illustrated, the casting **14** includes a front dash cross member **30**, a pair of opposed beams or inner rails **32***a*, **32***b*, a pair of opposed upper rails **34***a*, **34***b*, a pair of opposed aprons **36***a*, **36***b*, a plurality of mounting features **41** and a plurality of pockets **43***a*, **43***b* (together referred to as pockets **43**). A bumper (not shown) extends in a transverse direction Y relative to a longitudinal direction X of the vehicle **12** and is secured to front ends of the pair of beams **32***a*, **32***b*. The front dash cross member **30** extends in the transverse direction Y at a rear end **39***a* of the casting **14** and connects to the mid inner rails **32***a*, **32***b* and the subframe (not shown). Each beam **32***a*, **32***b* extends from a lower portion of a respective hinge pillar to the bumper. Each beam **32***a*, **32***b* is also arcuate and extends around a front wheel (not shown) of the vehicle **12** and forms a portion of a respective front wheel well. The upper rails **34***a*, **34***b* are positioned above the beams **32***a*, **32***b* and extend from an upper portion of a respective hinge pillar to a front end of a respective beam **32***a*, **32***b*. Stated differently, each upper rail **34***a*, **34***b* includes a first end **42***a* located at the upper portion of the respective hinge pillar and a second end **42***b* located at the front end of the respective beam **32***a*, **32***b*.

[0030] In the example illustrated, each apron **36***a*, **36***b* is arcuate and extends from a respective upper rail **34***a*, **34***b* to a respective beam **32***a*, **32***b*. In this way, each apron **36***a*, **36***b* is located between the respective upper rail **34***a*, **34***b* and the respective beam **32***a*, **32***b*. Each apron **36***a*, **36***b* also extends from the rear end **39***a* of the casting **14** to the front end **39***b* of the casting **14** and includes an opening **48** formed therein. That is, the opening **48** extends through each apron **36***a*, **36***b* such that one or more components (not shown) of the vehicle suspension system (not shown) may extend through (or pass through) the opening **48**. In the example illustrated, the opening **48** formed in the apron **36***a*, **36***b* is located closer toward a respective upper rail **34***a*, **34***b* than a respective beam **32***a*, **32***b*. In the example illustrated, each apron **36***a*, **36***b* includes a mounting feature or flange **52** that permits strut caps (not shown) to be mounted to the casting **14**. In the example illustrated, the mounting feature **52** extends upward from the apron **36***a*, **36***b* and extends at least partially around the opening **48** in the apron **36***a*, **36***b*.

[0031] Each strut cap (not shown) is secured to the mounting feature **52** of a respective apron **36***a*, **36***b* of the casting **14** and stores at least a portion of the vehicle suspension system. One example of such a strut cap is disclosed in the U.S. patent application Ser. No. ______, filed on the same day as the present application, and titled "VEHICLE STRUCTURAL ASSEMBLY HAVING STRUT CAPS," which is commonly owned with the present application and the contents of which are incorporated herein by reference in its entirety.

[0032] With reference to FIGS. **4**A, **4**B, and **6**, the plurality of mounting features **41** extend from the casting **14** and are attached to a portion of the vehicle **12**. For example, one or more mounting features **41** may be located in the area of the front dash cross member **30** and may secure the subframe (not shown) to the casting **14**. In another example, one or more mounting features **41** may be located in the areas of the aprons **36***a*, **36***b*, the upper rails **34***a*, **34***b*, and/or the beams **32***a*, **32***b* and may secure different parts or components of the vehicle **12** to the casting **14**. The mounting features **41** may have different lengths and/or thicknesses. As shown in FIG. **6**, each mounting feature **41** includes a proximal end **41***a* extending from the casting **14** and a distal end **41***b*. In some forms, the mounting feature **41** has a variable thickness from the proximal end **41***a* to the distal end **41***b*. In one example, the proximal end **41***a* has a thickness that is greater than a thickness of the distal end **41***a*. Stated differently, the mounting feature **41** is tapered from the proximal end **41***a* to the distal end **41***b*. In some forms, the mounting feature **41** has a constant thickness from the proximal end **41***a* to the distal end **41***b*. In the example illustrated, each mounting feature **41** is a boss that includes an aperture **58** extending at least partially therethrough and that is configured to receive a fastener (e.g., bolt, screw, and/or rivet). In some forms, the aperture **58** may be threaded.

[0033] With reference to FIGS. **2** and **3**, the pockets **43** are formed in the front dash cross member **30** of the casting **14**. In some forms, the pockets **43** may be formed at other areas of the casting **14** such as the aprons **36***a*, **36***b*, for example. In the example illustrated, the pockets **43** are formed in a wall **60** of the front dash cross member **30** and may extend in different directions. For example, one or more pockets **43***a* may have a length that extends along a longitudinal direction X of the vehicle **12** and/or one or more pockets **43***b* may have a length that extends at an angle relative to the longitudinal and transverse directions X, Y of the vehicle. As shown in FIG. **3**, pockets **43***b* extends in direction that is angled (e.g., acute angle) relative to the longitudinal and transverse directions X, Y. In the example illustrated, the wall **60** is angled or slanted. In some forms, the wall **60** is planar or flat.

[0034] With reference to FIGS. **5-7**, each pocket **43** has a non-circular shape and is recessed in-Z direction from the wall **60** of the front dash cross member **30**. Each pocket **43** is sized to receive one or more cooling dies during the casting process as will be described in more detail below. In the example illustrated, the pocket **43** has a generally Kammtail virtual foil profile and is defined by a base wall **62**, an arcuate wall **64**, and opposed side walls **66***a*, **66***b*. The base wall **62** is planar or flat and includes a first end **62***a* and a second end **62***b* that is opposite the first end **62***a*. The first end **62***a* is arcuate. In some forms, the second end **62***b* is straight. In other forms, the second end **62***b* is angled. One or more mounting features **41** extend in the −Z direction (FIG. **6**) from the base wall **62** such that the proximal ends **41***a* of the mounting features **41** are located proximate the base wall **62**. The mounting features **41** extending from the base wall **62** also extend in a direction that is perpendicular to the length of the pocket **43**. It should be understood that the aperture **58** in each mounting feature **41** does not extend through the base wall **62**.

[0035] The arcuate wall **64** extends in the +Z direction from the base wall **62** to the wall **60**. Stated different, the arcuate wall **64** extends from the base wall **62** in a direction that is opposite the direction the one or more mounting features **41** extend from the base wall **62**. The arcuate wall **64** extends from the side wall **66***a* of the pocket **43** to the side wall **66***b* of the pocket **43** that is opposite the side wall **66***a*. Stated differently, a first end **64***a* of the arcuate wall **64** is located at the side wall **66***a* of the pocket **43** and a second end **64***b* of the arcuate wall **64** is located at the side wall **66***b* of the pocket **43**. The side wall **66***a* extends from the first end **64***a* of the arcuate wall **64** to the second end **62***b* of the base wall **62** and the side wall **66***b* extends from the second end **64***b* of the arcuate wall **64** to the second end **62***b* of the base wall **62**. In some forms, the side walls **66***a*, **66***b* are straight. In other forms, the side walls **66***a*, **66***b* are curved outward (i.e., convex) from a center of the pocket **43**. A portion of the side wall **66***a* located proximate the arcuate wall **64** extends further in the +Z direction than a portion of the side wall **66***a* located proximate the second end **62***b* of the base wall **62**. Similarly, a portion of the side wall **66***b* located proximate the arcuate wall **64** extends further in the +Z direction than a portion of the side wall **66***b* located proximate the second end **62***b* of the base wall **62**.

[0036] With reference to FIGS. **8** and **9**, a method for manufacturing a part or structural casting of a vehicle **12** is provided. At **204**, molten (e.g., molten aluminum) is sent or directed through a cavity of a mold **94***a*, **94***b*. The mold **94***a*, **94***b* includes a mounting feature **96** and a cooling feature **97** adjacent to the mounting feature **96**. The cooling feature **97** is also shaped so as to reduce directional changes of the molten flowing through the cavity. For example, the cooling feature **97** may have a non-circular shape. In one example, the non-circular shape is a Kammtail virtual foil profile including an arcuate leading edge and a straight trailing edge. Molten flowing through the cavity flows substantially parallel to a longitudinal axis of the cooling feature **97**. The Kammtail virtual foil profile of the cooling feature **97** reduces turbulence in the wake of the cooling feature **97**. At **208**, one or more cooling dies **98** are inserted into a space of the mold **94***a*. In this way, molten received in the mounting feature **86** may be cooled, which reduces local temperatures and any shrink porosity. In the example illustrated, the cooling dies **98** are circular shaped and have cooling fluid (e.g., water) flowing therethrough. In some forms, the cooling dies **98** may have a different shape such as oval, rectangular, square, or any other suitable shape that can be received in the space of the mold **94***a*.

[0037] The casting **14** of the present disclosure provides pockets **43** including Kammtail virtual foil profiles, which reduces local hot spots and reduces air entrainment during manufacturing of the structural casting. Although the present disclosure discloses the casting **14** of the front-end structure of the vehicle **12** having pockets **43**, it should be understood that pockets **43** may be included in other parts of the vehicle **12** formed by a casting process such as hinge pillar reinforcements, mid rails, and kick down rails, for example.

[0038] Unless otherwise expressly indicated herein, all numerical values indicating mechanical/thermal properties, compositional percentages, dimensions and/or tolerances, or other characteristics are to be understood as modified by the word “about” or “approximately” in describing the scope of the present disclosure. This modification is desired for various reasons including industrial practice, material, manufacturing, and assembly tolerances, and testing capability.

[0039] As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

[0040] The description of the disclosure is merely exemplary in nature and, thus, variations that do not depart from the substance of the disclosure are intended to be within the scope of the disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the disclosure.

Claims

1. A structural assembly for a vehicle comprising: a cast body comprising a main wall, a non-circular shaped first pocket formed in the main wall, and at least one mounting feature, the first pocket defined at least partially by a base wall and an arcuate wall, the base wall having a first end and a second opposing end, the arcuate wall extending in a first direction from the first end of the base wall to the main wall, the arcuate wall also extending from a first side of the first pocket to a second opposing side of the first pocket, wherein the at least one mounting feature extends in a second direction from the base wall that is opposite the first direction, the at least one mounting feature is configured to be coupled to a vehicle component.
2. The structural assembly of claim 1, wherein the at least one mounting feature includes a plurality of mounting features extending in the second direction from the base wall.
3. The structural assembly of claim 1, wherein the at least one mounting feature is a boss, and wherein the boss comprises an aperture extending at least partially therethrough.
4. The structural assembly of claim 3, wherein the boss comprises a length having a variable thickness.
5. The structural assembly of claim 3, wherein the boss has a proximal end and a distal end, and wherein the proximal end has a greater thickness than a thickness of the distal end.
6. The structural assembly of claim 1, wherein the first pocket has a Kammtail virtual foil profile.
7. The structural assembly of claim 1, wherein the base wall is planar.
8. The structural assembly of claim 1, wherein the cast main body further includes a non-circular shaped second pocket, and wherein the second pocket has a Kammtail virtual foil profile.
9. The structural assembly of claim 8, wherein the first pocket extends in a third direction and the second pocket extends in a fourth direction, the third direction different from the fourth direction.
10. The structural assembly of claim 1, wherein the first pocket extends perpendicular to the at least one mounting feature.
11. The structural assembly of claim 1, wherein the cast main body is a front-end structure of the vehicle, and wherein the front end structure includes a left upper rail and a right upper rail.
12. A structural assembly for a vehicle comprising: a cast body comprising a main wall, a non-circular shaped first pocket formed in the main wall, and a plurality of bosses, the first pocket defined at least partially by a base wall and an arcuate wall, the base wall having a first end and a second opposing end that is straight, the arcuate wall extending in a first direction from the first end of the base wall to the main wall, the arcuate wall also extending from a first side of the first pocket to a second opposing side of the first pocket, wherein each boss of the plurality of bosses extend in a second direction from the base wall and comprises an aperture extending at least partially therethrough, the second direction being opposite the first direction, each boss of the plurality of bosses is configured to be coupled to a vehicle component.
13. The structural assembly of claim 12, wherein each boss has a proximal end and a distal end, and wherein the proximal end has a greater thickness than a thickness of the distal end.
14. The structural assembly of claim 12, wherein the first pocket has a Kammtail virtual foil profile.
15. The structural assembly of claim 12, wherein the cast body further includes a non-circular shaped second pocket, and wherein the second pocket has a Kammtail virtual foil profile.
16. The structural assembly of claim 15, wherein the first pocket extends in a third direction and the second pocket extends in a fourth direction, the third direction different from the fourth direction.
17. The structural assembly of claim 12, wherein the cast body is a front-end structure of the vehicle, and wherein the front end structure includes a left upper rail and a right upper rail.
18. The structural assembly of claim 12, wherein the main wall is inclined.

19. A structural assembly for a vehicle comprising: a cast body comprising a main wall, a plurality of non-circular shaped pockets formed in the main wall, and a plurality of bosses, each pocket is defined at least partially by a base wall and an arcuate wall, the base wall having a first end and a second opposing end that is straight, the arcuate wall extending in a first direction from the first end of the base wall to the main wall, the arcuate wall also extending from a first side of the first pocket to a second opposing side of the first pocket, wherein each boss of the plurality of bosses extend in a second direction from the base wall that is opposite the first direction, each boss is configured to be coupled to a vehicle component and comprises a proximal end and a distal end, the proximal end has a greater thickness than a thickness of the distal end, wherein a first non-circular shaped pocket of the plurality of non-circular shaped pockets extends in a third direction and a second non-circular shaped pocket of the plurality of non-circular shaped pockets extends in a fourth direction, the third direction different from the fourth direction.

20. The structural assembly of claim 19, wherein each non-circular shaped pocket of the plurality of non-circular shaped pockets has a Kammtail virtual foil profile.